



CUSTOMER BEHAVIOUR ANALYTICS

(Project Report)



BY
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1. INTRODUCTION

Customer data analysis plays an important role in understanding purchasing behaviour, customer preferences, and overall business performance. With the increasing availability of customer transaction data, data analytics techniques can be used to identify meaningful patterns and generate insights that support data-driven business decisions.

This project focuses on analysing customer purchase data containing 3,900 customer transaction records. The analysis examines customer demographics, purchasing behaviour, product categories, payment methods, shipping preferences, subscription status, discount usage, and customer ratings.

Exploratory data analysis techniques were used to identify patterns and trends in the dataset. In addition, K-Means clustering was applied to segment customers based on selected behavioural characteristics. An interactive dashboard was developed using Streamlit to present the analysis, key performance indicators, customer segments, and business insights in an accessible format.

2. PROBLEM STATEMENT

Businesses collect large amounts of customer purchase data, but raw transaction data alone does not provide meaningful information for decision-making. It is necessary to analyse customer characteristics and purchasing patterns to understand different customer groups, identify high-value customers, recognize areas requiring improvement, and develop appropriate business strategies.

The objective of this project is to analyse customer purchase data, identify significant patterns in customer behaviour, segment customers based on their characteristics and purchasing behaviour, and develop an interactive dashboard that communicates the findings effectively.

3. OBJECTIVES

- To perform data cleaning and preprocessing on the customer purchase dataset.
- To analyse customer demographics and purchasing behaviour.
- To identify product categories and items with higher purchase activity.
- To analyse payment methods and shipping preferences.
- To study subscription and discount usage patterns.
- To examine customer ratings and purchasing behaviour.
- To segment customers using the K-Means clustering algorithm.
- To develop an interactive dashboard using *Streamlit*.
- To generate meaningful business insights and recommendations from the analysis.

4. DATASET DESCRIPTION

The dataset used for this project is a publicly available customer shopping dataset containing 3,900 customer purchase records and 18 original attributes. The dataset provides information about customer demographics, purchased products, purchasing behaviour, payment preferences, subscription status, discounts, customer ratings, and previous purchase history.

The dataset was used to perform exploratory data analysis, identify purchasing patterns, and segment customers based on selected behavioural characteristics. Additional derived attributes were created during the analysis to support customer profiling and segmentation.

Dataset dimensions

Property	Value
Number of records	3,900
Number of attributes (Original & Final)	18 & 22
Dataset type	Customer purchase/ transaction data
Analysis performed	EDA and customer segmentation
Segmentation technique	K-Means Clustering
Dashboard tool	Streamlit

4.1 Data Dictionary

Column	Description
Customer ID	Unique identifier associated with each customer
Age	Age of the customer
Gender	Gender of the customer
Item Purchased	Product/item purchased by the customer
Category	Category of the purchased product
Purchase Amount (USD)	Amount spent on the purchase in USD
Location	Customer's location
Size	Size selected for the purchased product
Color	Colour of the purchased product
Season	Season associated with the purchase
Review Rating	Rating provided by the customer
Subscription Status	Indicates whether the customer has a subscription
Shipping Type	Shipping method selected for the purchase
Discount Applied	Indicates whether a discount was applied
Promo Code Used	Indicates whether a promotional code was used
Previous Purchases	Number of previous purchases made by the customer
Payment Method	Payment method used for the purchase
Frequency of Purchases	Frequency with which the customer makes purchases

4.2 Tools and Technologies

Tools/ Technology	Purpose
Python	Data analysis and preprocessing
Pandas	Data manipulation and analysis
Numpy	Numerical operations
Matplotlib	Data visualizations
Seaborn	Statistical visualizations
Scikit-learn	K-Means customer segmentation
Google Colab	Data analysis and experimentation
Streamlit	Interactive dashboard development
Microsoft Word	Project report preparation

5. DATA PREPROCESSING

The original customer shopping dataset contained 3,900 records and 18 attributes. An initial inspection of the dataset was performed to understand its structure, data types, and completeness. The dataset contained **no missing/ null values**, so no missing-value imputation or deletion was required.

The numerical and categorical attributes were examined separately to support subsequent exploratory data analysis. Additional analytical features, including **Age Group, Spending Level, and Loyalty Level**, were derived to facilitate customer profiling. The prepared dataset was subsequently used for exploratory analysis and customer segmentation using the K-Means clustering algorithm.

6. FEATURE ENGINEERING

To improve the analysis of customer characteristics and purchasing behaviour, three additional categorical features were derived from the existing numerical attributes: **Age Group, Spending Level, and Loyalty Level**. These features were created to make customer profiling easier and to support the subsequent customer segmentation analysis.

6.1 Age Group

The **Age** attribute was converted into six age categories:

Age Range	Age Group
18-25	18-25
26-35	26-35
36-45	36-45
46-55	46-55
56-65	56-65
65+	65+

The age groups were created using predefined age intervals. This transformation makes it easier to compare purchasing behaviour across different age categories.

6.2 Spending Level

The **Purchase Amount (USD)** variable was divided into **3 groups using quantile-based binning** (qcut):

Spending Level	Description
Low	Lower third of purchase amounts
Medium	Middle third of purchase amounts
High	Upper third of purchase amounts

Quantile-based categorization was used so that the purchase amounts were divided according to their distribution in the dataset.

6.3 Loyalty Level

The **Previous Purchases** attribute was categorised into 4 loyalty levels:

Previous Purchases	Loyalty Level
0-15	New
16-30	Regular
31-45	Loyal
46-60	VIP

The loyalty levels were derived from the number of previous purchases. Customers with fewer previous purchases were categorized as New or Regular, while customers with higher previous purchase counts were categorized as Loyal or VIP.

Summary Table

Derived Feature	Source Attribute	Method
Age Group	Age	pd.cut()
Spending Level	Purchase Amount (USD)	pd.qcut(q=3)
Loyalty Level		pd.cut()

7. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics and purchasing behaviour of customers in the dataset. Various demographic, product, purchasing, payment, subscription, and customer satisfaction attributes were analysed using statistical summaries and visualizations. The analysis helped identify important patterns and trends that were subsequently used for customer segmentation and business interpretation.

7.1 Customer Demographic Analysis

Customer demographic characteristics were analysed to understand the composition of the customer base. Gender and age-related attributes were examined to identify the distribution of customers across different demographic groups. The dataset contains 3,900 customer records with male customers for 68% of the dataset and female customers accounting for 32%. The age distribution was further analysed by grouping customers into six age categories.

Gender Distribution

Gender	Number of Customers	Percentage
Male	2,652	68%
Female	1,248	32%
Total	3,900	100%

Observation: The customer dataset is predominantly male, with male customers representing 68% of the records, while female customers account for 32%.

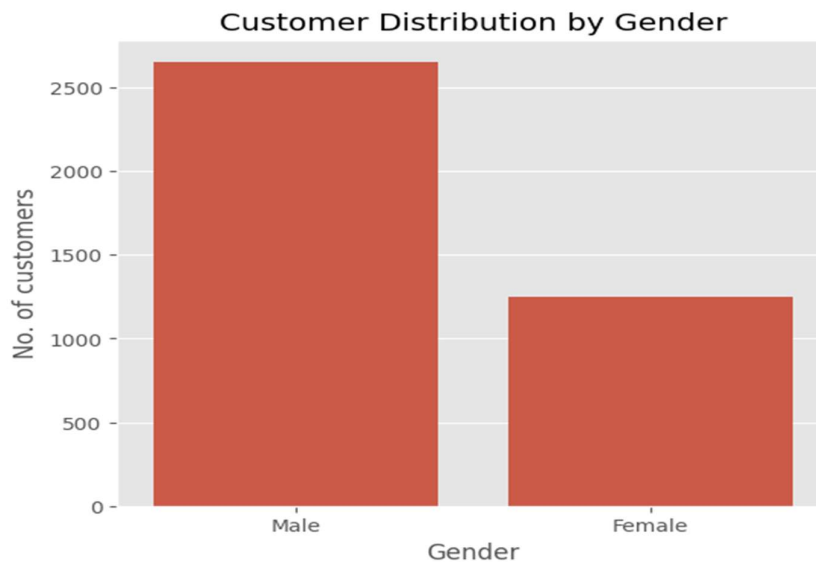


Figure 1: Gender Distribution of Customers

Age Group Distribution

Age Group	Number of Customers
18-25	571
26-35	742
36-45	729
46-55	753
56-65	750
65+	355
Total	3,900

Observation: The age-group analysis shows that the largest customer group is the 46–55 age group, with 753 customers, followed closely by the 56–65 age group, with 750 customers. The 26–35 and 36–45 age groups contain 742 and 729 customers, respectively. The 65+ group has the lowest representation with 355 customers.

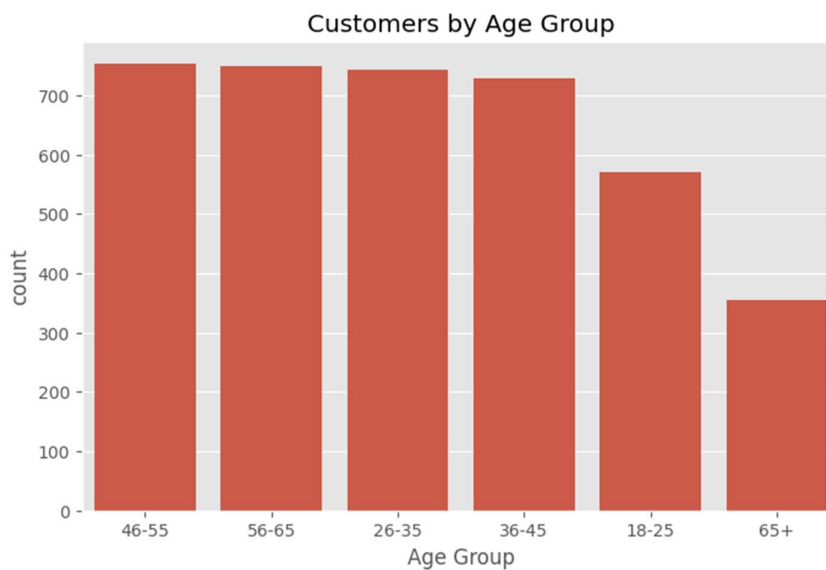


Figure 2: Customer Distribution by Age Group

7.2 Product and Category Analysis

Product-level analysis was performed to understand customer preferences and evaluate the contribution of different product categories to overall revenue. The number of purchases, total revenue, and average purchase amount were calculated for each category. In addition, the most frequently purchased items and the highest-revenue items were identified.

7.2.1 Category-wise Analysis

Category	Number of Purchases	Total Revenue (USD)	Average Purchase (USD)
Clothing	1,737	104,264	60.03
Accessories	1,240	74,200	59.84
Footwear	599	36,093	60.26
Outerwear	324	18,524	57.17
Total	3,900	233,081	--

Observation: Among the four product categories, Clothing recorded the highest number of purchases, with 1,737 purchases, and generated the highest total revenue of \$104,264. Accessories ranked second with 1,240 purchases and revenue of \$74,200. Footwear generated \$36,093 from 599 purchases, while Outerwear had the lowest purchase count of 324 and revenue of \$18,524.

The average purchase amount was relatively similar across the categories, ranging from approximately **\$57.17 to \$60.26**, indicating that the major difference in total revenue is largely associated with the number of purchases rather than a very large difference in average transaction value.

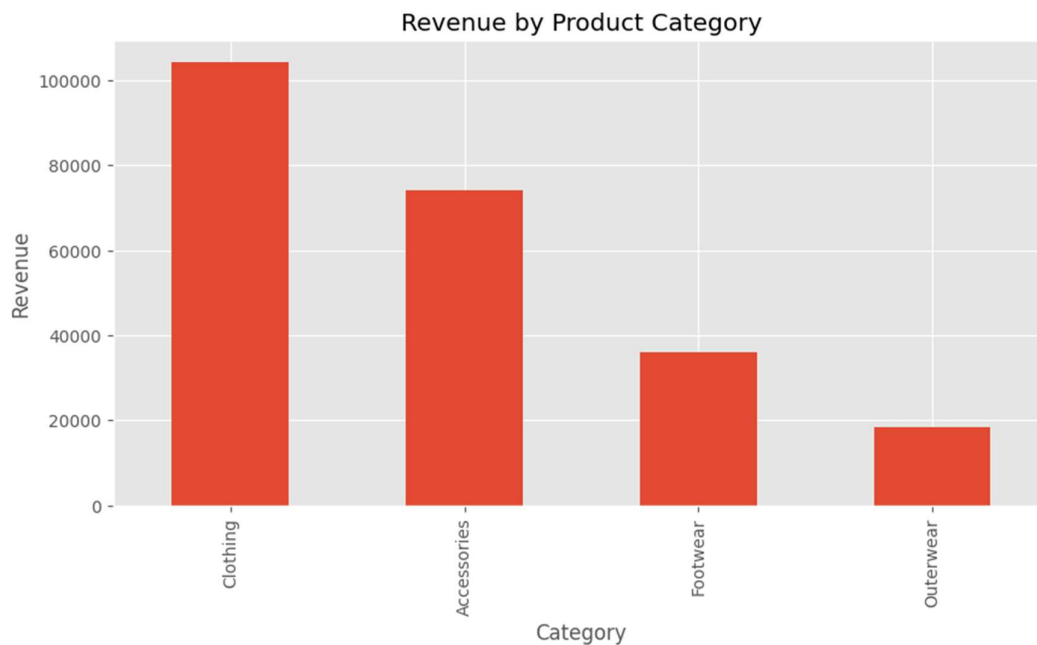


Figure 3: Category-wise number of purchases and revenue

7.2.2 Top 10 Purchased Items

Rank	Item	Number of Purchases
1	Blouse	171
2	Pants	171
3	Jewellery	171
4	Shirt	169
5	Dress	166
6	Sweater	164
7	Jacket	163
8	Coat	161
9	Sunglasses	161
10	Belt	161

Observation: Blouse, Pants, and Jewellery were the most frequently purchased individual items, each recording **171 purchases**. Shirt followed with 169 purchases, while Dress recorded 166 purchases. The remaining items in the top 10 had purchase counts between 161 and 164.

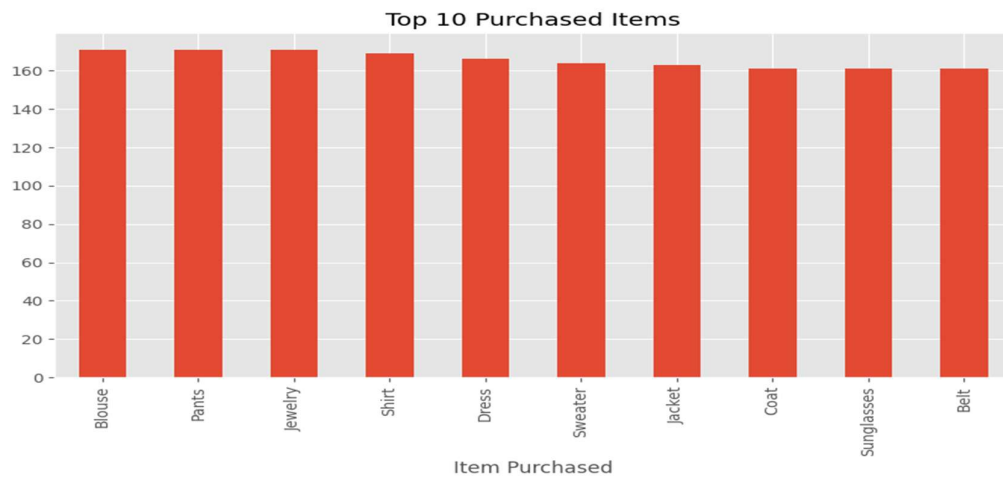


Figure 4: Top 10 Purchased Items

7.2.3 Top 10 Items by Revenue

Rank	Item	Total Revenue (USD)
1	Blouse	10,410
2	Shirt	10,332
3	Dress	10,320
4	Pants	10,090
5	Jewellery	10,010
6	Sunglasses	9,649
7	Belt	9,635
8	Scarf	9,561
9	Sweater	9,462
10	Shorts	9,433

Observation: Blouse generated the highest revenue among individual products at \$10,410, followed by Shirt at \$10,332 and Dress at \$10,320. The revenue values among the top 10 products were relatively close, ranging from \$9,433 to \$10,410.

7.3 Payment and Shipping Analysis

Payment and shipping preferences were analysed to understand the methods customers most commonly used for completing and receiving their purchases. The **Payment Method** and **Shipping Type** attributes were examined using purchase-frequency counts.

7.3.1 Payment Method Analysis

Payment Method	Number of Purchases
PayPal	677
Credit Card	671
Cash	670
Debit Card	636
Venmo	634
Bank Transfer	612
Total	3,900

Observation: PayPal was the most frequently used payment method, accounting for 677 purchases, followed by Credit Card with 671 and Cash with 670 purchases. Bank Transfer was the least frequently used method, with 612 purchases. Overall, the differences between payment methods were relatively small, indicating that customers used a variety of payment options.

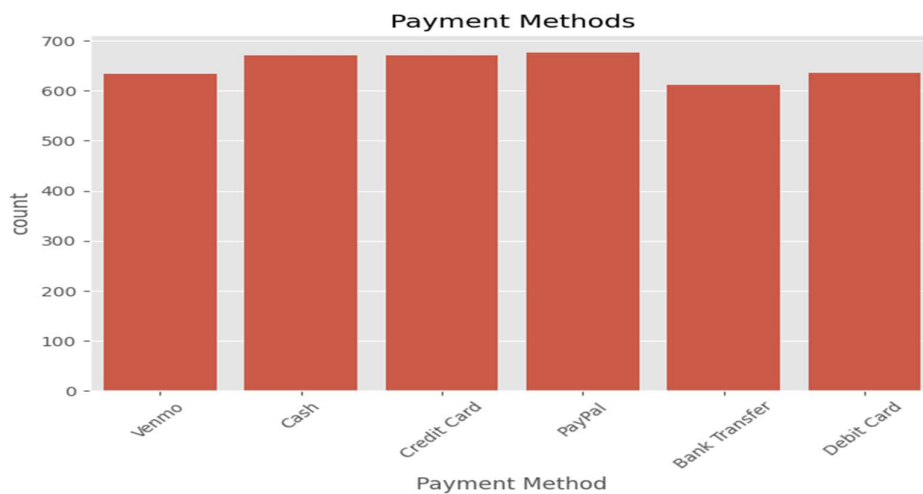


Figure 5: Customer Purchases by Payment Method

7.3.2 Shipping Type Analysis

Shipping Type	Number of Purchases
Free Shipping	675
Standard	654
Store Pickup	650
Next Day Air	648
Express	646
2-Day Shipping	627
Total	3,900

Observation: Free Shipping was the most frequently selected shipping option, with 675 purchases, while 2-Day Shipping had the lowest frequency at 627 purchases. The relatively narrow difference between the shipping options suggests that customers used a variety of delivery methods rather than strongly preferring a single option.



Figure 6: Customer Purchases by Shipping Type

7.4 Subscription and Discount Analysis

Subscription and promotional behaviour were analysed to examine the relationship between customer subscriptions, discounts, and purchasing activity. The analysis also compared the purchasing behaviour of subscribed and non-subscribed customers using total revenue and average purchase amount.

7.4.1 Subscription Status

Subscription Status	Customers	Percentage
No	2,847	73%
Yes	1,053	27%
Total	3,900	100%

Subscription Purchase Analysis

Subscription Status	Customers	Total Revenue (USD)	Average Purchase (USD)
No	2,847	170,436	59.87
Yes	1,053	62,645	59.49

Observation: The majority of customers, **2,847 (73%)**, were not subscribed, while **1,053 customers (27%)** had an active subscription. Non-subscribed customers generated higher total revenue of **\$170,436**, compared with \$62,645 from subscribed customers. However, the average purchase amounts were very similar, at **\$59.87 for non-subscribed customers and \$59.49 for subscribed customers**. The similar average purchase amounts indicate that subscription status did not produce a substantial difference in the average transaction value within this dataset.

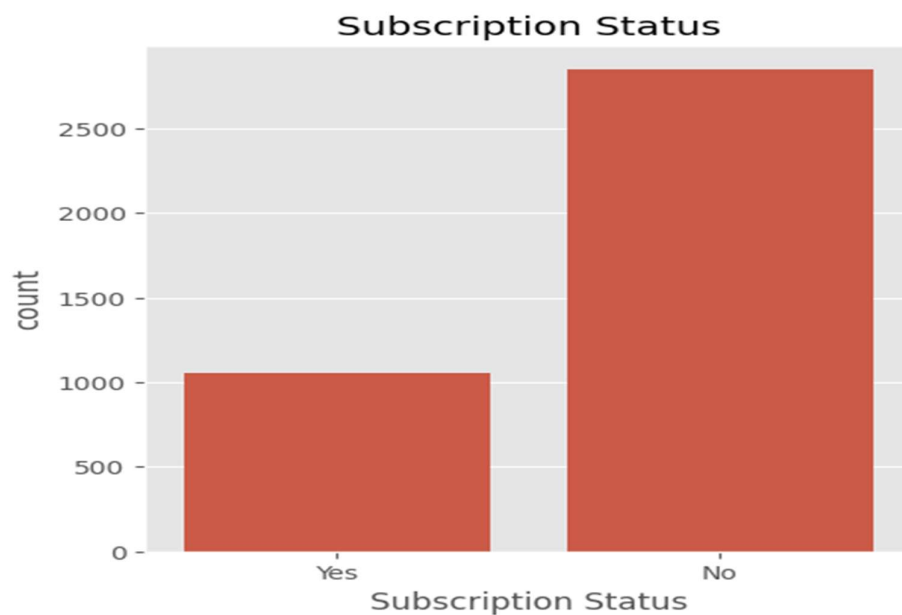


Figure 7: Subscription status Distribution

7.4.2 Discount Analysis

Discount Applied	Number of Purchases	Percentage
No	2,223	57%
Yes	1,677	43%
Total	3,900	100%

Observation: Discounts were applied to 1,677 purchases (43%), while 2,223 purchases (57%) were made without a discount. This indicates that discount usage was relatively common, although purchases without discounts represented the larger share of transactions.

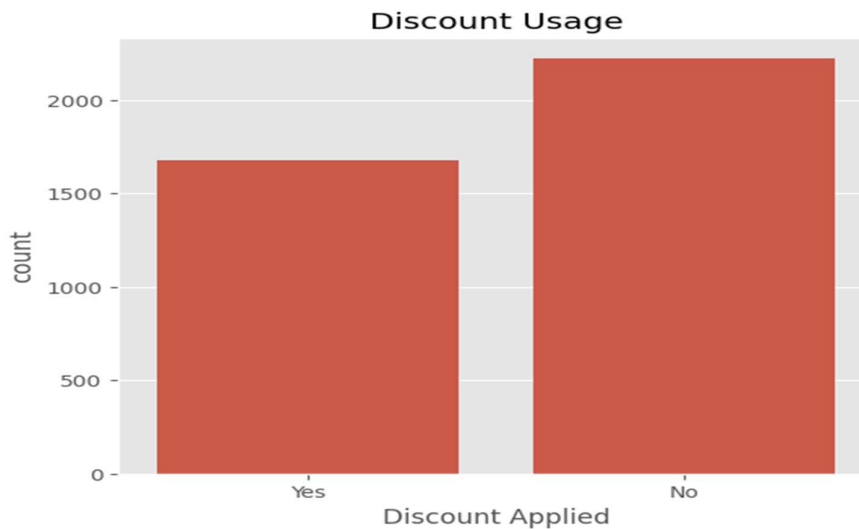


Figure 8: Discount Usage Distribution

7.4.3 Promotional Code Usage

Promo Code Used	Number of Purchases	Percentage
No	2,223	57%
Yes	1,677	43%
Total	3,900	100%

Observation: Promotional code usage followed the same distribution as discount application, with promotional codes used in 1,677 purchases (43%) and not used in 2,223 purchases (57%). The identical distributions of the Discount Applied and Promo Code Used attributes suggest that these two variables have the same recorded pattern in the dataset.

7.5 Purchase Behaviour Analysis

Customer purchasing behaviour was analysed using purchase amount, previous purchases, purchase frequency, and the derived Spending Level and Loyalty Level attributes. These variables provided insights into the differences between customers based on their current and historical purchasing activity.

7.5.1 Purchase Amount Analysis

Statistic	Purchase Amount (USD)
Count	3,900
Mean	59.76
Standard Deviation	23.69
Minimum	20.00
25 th Percentile	39.00
Median	60.00
75 th Percentile	81.00
Maximum	100.00

Observation: The average purchase amount was \$59.76, while the median purchase amount was \$60.00, indicating that the centre of the purchase distribution is around \$60. Purchase amounts ranged from \$20 to \$100, with a standard deviation of approximately \$23.69. The middle 50% of purchases were between \$39 and \$81.

7.5.2 Previous Purchase Analysis

Statistic	Previous Purchases
Count	3,900
Mean	25.35
Standard Deviation	14.45
Minimum	1
25 th Percentile	13
Median	25
75 th Percentile	38
Maximum	50

Observation: Customers had an average of 25.35 previous purchases, with a median of 25. The number of previous purchases ranged from 1 to 50. The middle 50% of customers had between 13 and 38 previous purchases, indicating considerable variation in customers' purchase histories.

7.5.3 Purchase Frequency

Frequency	Customers
Every 3 months	584
Annually	572
Quarterly	563
Monthly	553
Bi-Weekly	547
Fortnightly	542
Weekly	539
Total	3,900

Observation: The most common reported purchase frequency was **Every 3 Months**, with 584 customers, followed by Annual purchases with 572 customers. Weekly purchasing had the lowest representation, with 539 customers. The relatively close counts across the different frequency categories indicate that customers are distributed across a wide range of purchasing frequencies.

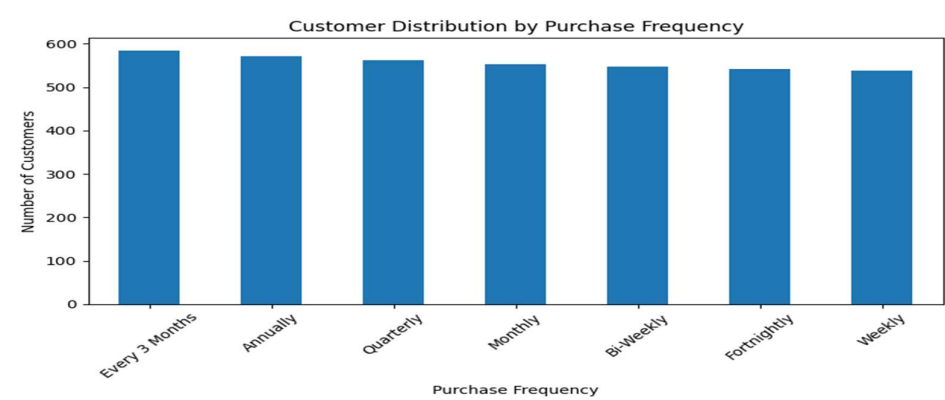


Figure 9: Customer Distribution by Purchase Frequency

7.5.4 Spending Level

Spending Level	Customers	Percentage
Low	1,300	33.33%
Medium	1,303	33.41%
High	1,297	33.26%
Total	3,900	100%

Observation: Customers were divided into Low, Medium, and High spending groups using quantile-based categorization. The three groups contain approximately one-third of the customer records each, with 1,300 Low-spending, 1,303 Medium-spending, and 1,297 High-spending customers.

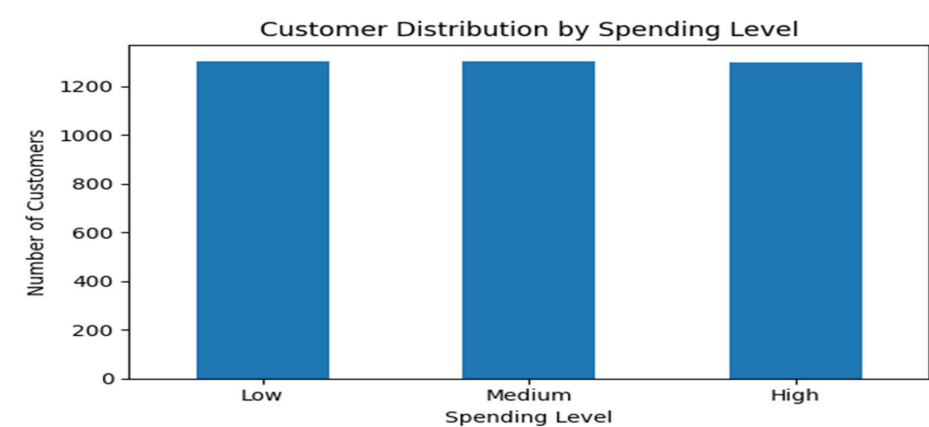


Figure 10: Customer Distribution by Spending Level

7.5.5 Loyalty Level

Loyalty Level	Customers	Percentage
New	1,179	30.23%
Regular	1,172	30.05%
Loyal	1,156	29.64%
VIP	393	10.08%
Total	3,900	100%

Observation: Based on the number of previous purchases, **1,179 customers were classified as New**, 1,172 as Regular, 1,156 as Loyal, and 393 as VIP. The New, Regular, and Loyal groups had relatively similar representation, while the VIP segment represented approximately **10%** of the dataset.

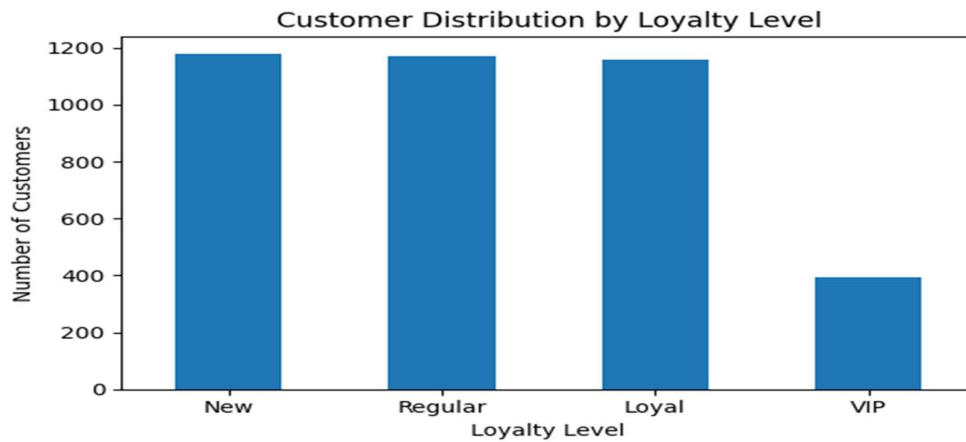


Figure 11: Loyalty Level

Customer Rating Analysis

Customer ratings were analysed to understand customer satisfaction patterns and to examine differences in ratings across customer groups. The Review Rating attribute was examined using descriptive statistics and frequency distribution to identify the central tendency and variation in customer ratings.

Review Rating Statistics

Statistic	Review Rating
Number of ratings	3,900
Mean	3.75
Standard Deviation	0.72
Minimum	2.50
25 th Percentile	3.10
Median	3.70
75 th Percentile	4.40

Maximum	5.00
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Observation: The average customer rating was approximately 3.75 out of 5, with a median rating of 3.70. Ratings ranged from 2.5 to 5.0, with a standard deviation of approximately 0.72. The middle 50% of ratings fell between 3.1 and 4.4, indicating a moderate spread in customer feedback. The distribution covers a broad range of ratings, although the average rating is above the midpoint of the available rating range.

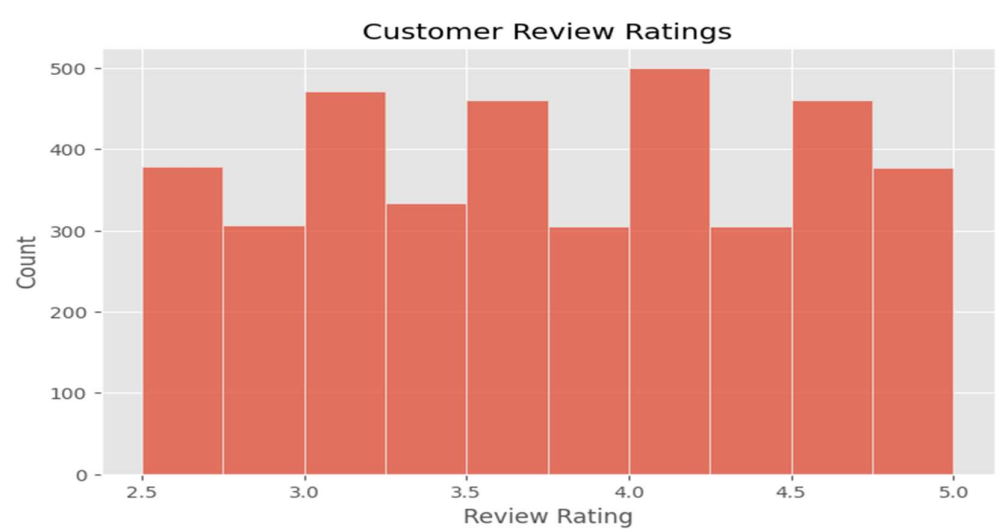


Figure 12: Distribution of Customer Review Ratings

8. CUSTOMER SEGMENTATION

Customer segmentation was performed to divide customers into groups with similar characteristics and purchasing behaviour. The K-Means clustering algorithm was selected to identify distinct customer groups based on numerical customer attributes. This approach helps in understanding different customer profiles and can support targeted business strategies.

8.1 Features Used for Clustering

Four numerical attributes were selected as input features for the K-Means clustering algorithm. These features represent customer demographics, purchasing behavior, historical purchase activity, and customer satisfaction.

Feature	Description	Role in Segmentation
Age	Customer’s age	Represents demographic characteristics
Purchase Amount (USD)	Amount spent on the purchase	Represents spending behaviour

Previous Purchases	Number of previous purchases	Represents purchase history and customer loyalty
Review Rating	Customer's rating	Represents customer feedback/satisfaction

8.2 Data Standardization

Before applying the K-Means clustering algorithm, the selected numerical features were standardized using *StandardScaler* from the Scikit-learn library. Standardization was performed to transform the features to a comparable scale, preventing variables with larger numerical ranges, such as Purchase Amount and Previous Purchases, from disproportionately influencing the clustering process. The four selected features—Age, Purchase Amount (USD), Previous Purchases, and Review Rating—were transformed using standardization before being provided to the K-Means algorithm.

8.3 Determining the Number of Clusters

The Elbow Method was used to determine the appropriate number of clusters for the K-Means algorithm. The Within-Cluster Sum of Squares (WCSS) was calculated for values of K ranging from 1 to 10. The WCSS decreases as the number of clusters increases; however, the rate of decrease becomes less pronounced after $K = 4$. Therefore, **$K = 4$ was selected as the appropriate number of clusters** for the customer segmentation analysis.

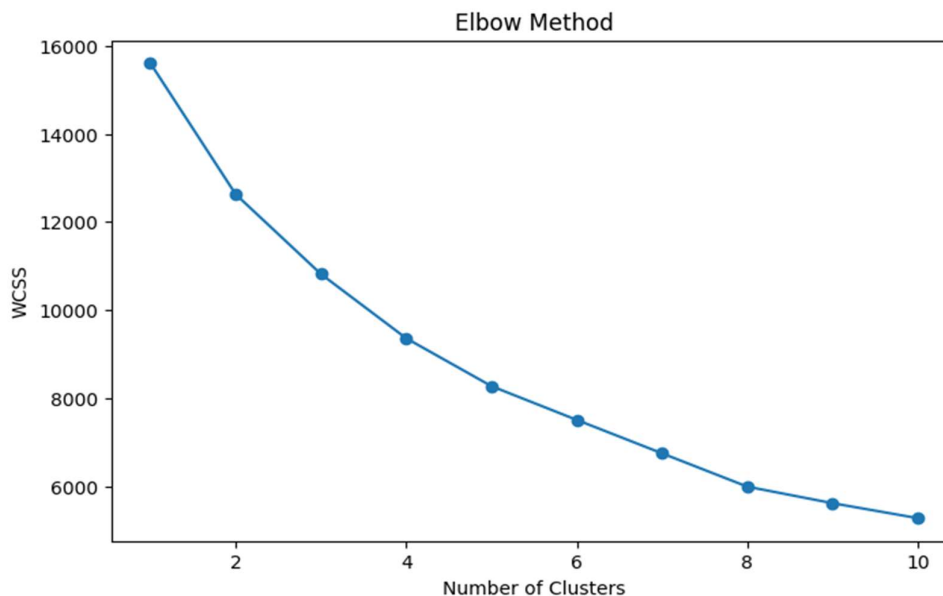


Figure 13: Elbow Method: $K = 4$ was selected based on the point where the WCSS reduction begins to become less pronounced.

8.4 Applying K-Means Clustering

After selecting four clusters using the Elbow Method, the K-Means clustering algorithm was applied to the standardized customer features. The model was configured with **four clusters** ($K = 4$), a *random_state* of 42 for reproducibility, and *n_init* = 10 to perform multiple centroid initializations. The fitted model assigned each customer to one of the four clusters, and the resulting cluster labels were stored in a new Cluster column.

8.5 Cluster Distribution

After applying K-Means clustering, the 3,900 customers were divided into four clusters. The distribution of customers across the clusters was relatively balanced, with each cluster representing approximately one-quarter of the total customer base.

Cluster	Number of Customers	Percentage
Cluster 0	966	24.77%
Cluster 1	969	24.85%
Cluster 2	986	25.28%
Cluster 3	979	25.10%
Total	3,900	100%

Observation: Cluster 2 contains the largest number of customers, with **986 customers (25.28%)**, while Cluster 0 contains the smallest group, with **966 customers (24.77%)**. The difference between the largest and smallest clusters is only 20 customers, indicating a relatively balanced segmentation.



Figure 14: Customer Segments

8.6 Cluster Profiling

Cluster	Customers	Avg. Age	Avg. Purchase	Avg. Previous Purchases	Avg. Rating
0	966	56.22	82.08	26.76	3.96
1	969	31.33	54.86	24.98	4.43
2	986	32.59	64.76	22.44	3.09
3	979	56.26	37.57	27.26	3.54

Cluster	Proposed Segmented Name	Main Characteristics
0	High-Value Mature Customers	Older, highest purchase amount, relatively good ratings
1	Young Satisfied Customers	Younger, moderate spending, highest ratings
2	Young Moderate Spending Customers	Younger, higher spending than Cluster 1, lowest ratings
3	Mature Low – Spending Customers	Older, lowest spending, highest previous purchases

8.7 Business Interpretation of Customer Segments

Cluster 0 --- High-Value Mature Customers

Profile:

- 966 Customers
- Average age: 56.22
- Average Purchase: \$82.08 ---- highest
- Previous Purchases: 26.76
- Rating: 3.96

Interpretation:

Cluster 0 represents high-value mature customers who generate the highest average purchase amount among all four segments. Their relatively high number of previous purchases also indicates an established purchasing relationship with the business.

Possible Business Strategy:

The business can focus on customer retention through personalized offers, premium product recommendations, loyalty rewards, and early access to new products.

Cluster 1 --- Young Satisfied Customers

Profile:

- 969 customers
- Average age: 31.33
- Average purchase: \$54.86
- Previous purchases: 24.98
- Rating: 4.43 --- highest

Interpretation:

Cluster 1 consists of younger customers with moderate spending levels and the highest average review rating. The high rating suggests relatively positive customer feedback from this segment.

Possible Business Strategy:

The business can focus on maintaining customer satisfaction and encouraging repeat purchases through personalized recommendations, loyalty programs, and targeted promotional campaigns.

Cluster 2 --- Young Moderate Spending Customers

Profile:

- 986 customers --- largest cluster
- Average age: 32.59
- Average purchase: \$64.76
- Previous purchases: 22.44
- Rating: 3.09 --- lowest

Interpretation:

Cluster 2 represents younger customers who have relatively higher average purchase values but the lowest average review rating among the four segments. This combination makes the segment important for further investigation because relatively strong spending is accompanied by comparatively lower customer ratings.

Possible Business Strategy:

The business could analyse customer feedback from this group to identify potential areas for improvement. Personalized engagement and product/service improvements may help increase satisfaction while retaining their existing purchasing activity.

Cluster 3 --- Mature Low Spending Customers

Profile:

- 979 customers
- Average age: 56.26
- Average purchase: \$37.57 --- lowest
- Previous purchases: 27.26 --- highest
- Rating: 3.54

Interpretation:

Cluster 3 consists of mature customers with the highest average number of previous purchases but the lowest average current purchase amount.

Possible Business Strategy:

The business could use personalized product recommendations, cross-selling, bundle offers, and loyalty incentives to encourage these customers to increase their purchase value.

9. INTERACTIVE DASHBOARD DEVELOPMENT

An interactive customer analytics dashboard was developed using **Streamlit** to present the results of the customer data analysis in an intuitive and interactive format. The dashboard combines key performance indicators, customer demographics, product analysis, purchasing behaviour, and customer segmentation in a single interface. Interactive filters allow users to explore the data based on selected customer attributes.

9.1 KPI Cards

Four key performance indicators (KPIs) were included at the top of the dashboard to provide a quick overview of the customer dataset and overall purchasing performance.

KPI	Value
Total Customers	3,900
Total Revenue	\$233,081
Average Purchase	\$59.76
Average Rating	3.75

Description:

- **Total Customers:** Displays the total number of customer purchase records available in the dataset.
- **Total Revenue:** Represents the total purchase revenue generated across all 3,900 records.
- **Average Purchase:** Shows the average purchase amount across all customer transactions.

- **Average Rating:** Represents the average customer review rating recorded in the dataset.

9.2 Customer Overview

The Customer Overview section provides a visual summary of the customer demographic composition. The dashboard includes visualizations for gender distribution and age-group distribution, allowing users to examine the characteristics of the customer base interactively.

9.3 Product Analysis

The Product Analysis section focuses on purchasing patterns across different product categories and individual items. A revenue-by-category visualization was included to compare the contribution of each product category to overall revenue. A second visualization displays the top 10 most frequently purchased items.

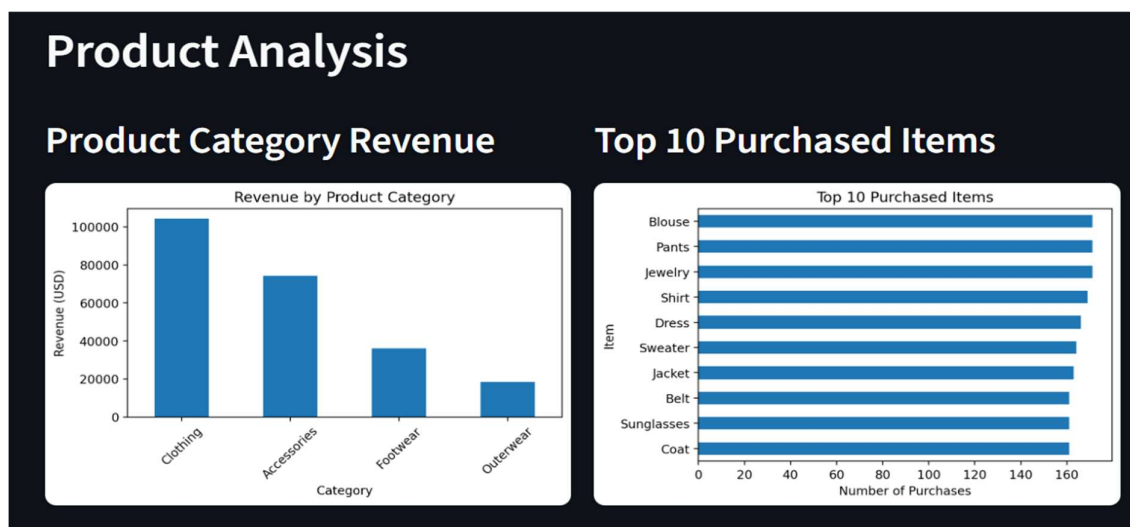


Figure 15: Product Analysis from Dashboard

9.4 Customer Segmentation

This section presents the results obtained from the K-Means clustering analysis. The dashboard displays the distribution of customers across the four identified clusters and provides the major characteristics of each customer segment. Descriptive segment names were assigned to the clusters based on their average age, purchase amount, previous purchases, and review rating.

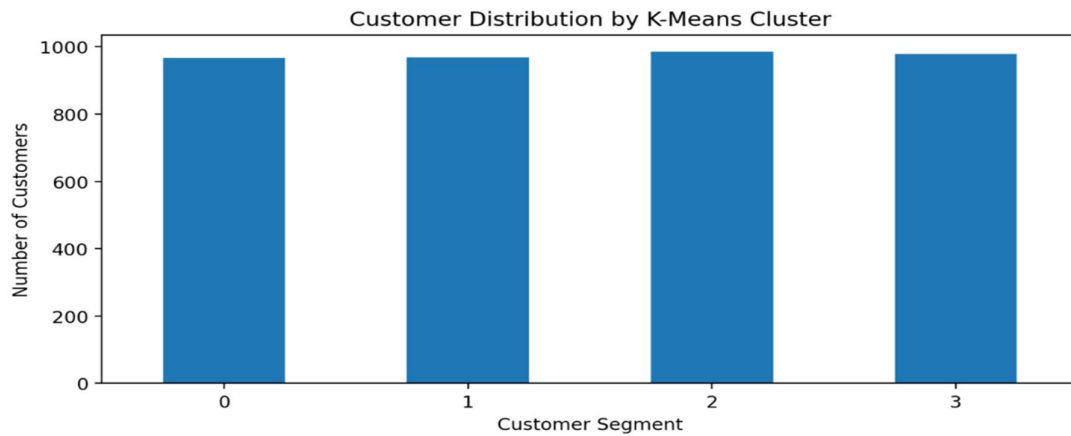


Figure 16: Cluster Distribution

9.5 Interactive Filters

Interactive filters were incorporated into the dashboard to allow users to analyze specific subsets of customers. Users can modify the selected criteria and observe the corresponding changes in the displayed visualizations. This makes the dashboard useful for exploring customer behaviour from different perspectives.



Figure 17: Apply Filter

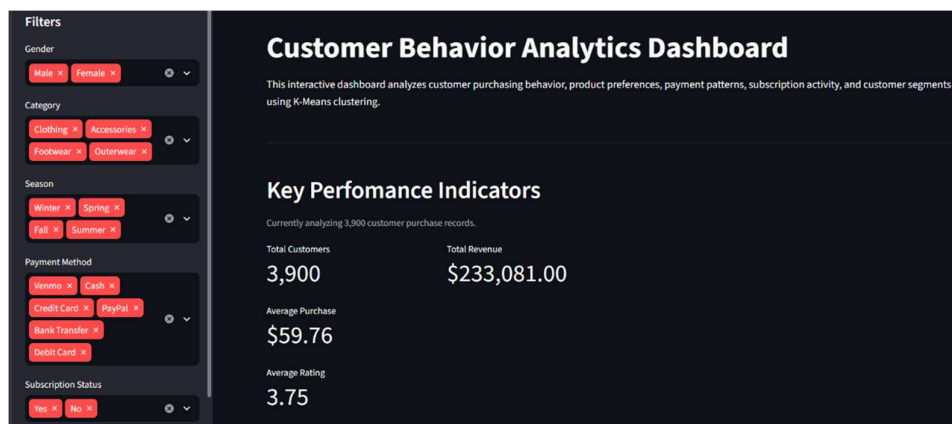


Figure 18: No Filter

10. KEY FINDINGS

The customer data analysis provided several important insights into customer demographics, purchasing behaviour, product performance, and customer segmentation. The major findings obtained from the analysis are summarized below.

10.1 Customer Demographics

The dataset contains 3,900 customer purchase records, of which **68% are male customers and 32% are female customers**. The largest age group is **46–55 years**, with 753 customers, followed closely by the 56–65 age group with 750 customers.

10.2 Product Performance

Clothing was the strongest-performing product category, recording 1,737 purchases and generating the highest category revenue of **\$104,264**. The overall purchase revenue across the dataset was **\$233,081**. Among individual products, Blouse recorded the highest total revenue at **\$10,410**.

10.3 Payment and Shipping Preferences

Customer preferences were relatively evenly distributed across the available payment and shipping methods. **PayPal** was the most frequently used payment method with 677 purchases, while **Free Shipping** was the most frequently selected shipping option with 675 purchases.

10.4 Subscription and Discount Behaviour

Non-subscribed customers represented **73%** of the dataset, while subscribed customers accounted for 27%. The average purchase amount was similar for both groups, at **\$59.87 for non-subscribed customers and \$59.49 for subscribed customers**. Discounts were applied to 43% of purchases, while 57% were made without discounts.

10.5 Purchasing Behaviour

The average purchase amount was **\$59.76**, with individual purchase amounts ranging from \$20 to \$100. Customers had an average of **25.35 previous purchases**. The Spending Level categories were approximately equally distributed because they were created using quantile-based binning.

10.6 Customer Satisfaction

The average review rating was **3.75 out of 5**, with ratings ranging from 2.5 to 5.0. The median rating was 3.70, indicating that customer ratings were generally centred around the middle-to-upper portion of the rating range.

10.7 Customer Segmentation

K-Means clustering divided the customers into four relatively balanced segments. **Cluster 0** represented high-value mature customers with the highest average purchase amount of **\$82.08**. **Cluster 1** consisted of younger customers with the highest average rating of **4.43**. **Cluster 2** was the largest cluster and had the lowest average rating of **3.09**, while **Cluster 3** had the lowest average purchase amount of **\$37.57** but the highest average number of previous purchases at **27.26**.

11. BUSINESS RECOMMENDATIONS

11.1 Retain High-Value Customers

High-value mature customers in Cluster 0 have the highest average purchase amount. The business could focus on retaining this segment through personalized recommendations, loyalty rewards, premium product offers, and exclusive promotions.

11.2 Improve Customer Satisfaction in Cluster 2

Cluster 2 has relatively high average spending but the lowest average review rating. The business should investigate customer feedback from this segment to identify possible issues affecting satisfaction and develop targeted improvements.

11.3 Encourage Higher Spending Among Cluster 3

Cluster 3 has the highest average number of previous purchases but the lowest average purchase amount. Cross-selling, product bundles, and personalized recommendations could potentially encourage customers in this segment to increase their transaction value.

11.4 Strengthen Engagement with Younger Customers

Cluster 1 consists of younger customers with the highest average rating. Maintaining their positive experience through personalized communication, loyalty programs, and relevant product recommendations could help encourage continued engagement.

11.5 Monitor Discount and Promotional Strategies

Discounts and promotional codes were used in 43% of purchases. The business can monitor the effectiveness of promotional campaigns by comparing purchase behaviour and revenue generated from discounted and non-discounted transactions.

12. CONCLUSION

This project analysed 3,900 customer purchase records to identify patterns in customer demographics, product preferences, purchasing behaviour, payment and shipping choices, subscription activity, and customer satisfaction. Exploratory Data Analysis was used to identify important patterns, while additional features such as Age Group, Spending Level, and Loyalty Level were created to support customer profiling.

K-Means clustering was subsequently applied using Age, Purchase Amount, Previous Purchases, and Review Rating to segment customers into four distinct groups. The resulting cluster profiles revealed differences in customer age, spending behaviour, purchase history, and review ratings. An interactive Streamlit dashboard was developed to present these findings through KPI cards, visualizations, filters, and customer segment profiles.

Overall, the analysis demonstrates how customer data analytics and unsupervised machine learning can be used to understand different customer groups and support data-driven business strategies.

13. LIMITATIONS & FUTURE SCOPE

The analysis has certain limitations based on the available dataset. The dataset does not contain a transaction date field; therefore, detailed time-based sales trend analysis such as monthly or yearly revenue trends could not be performed. The customer segmentation was based on four numerical attributes—Age, Purchase Amount, Previous Purchases, and Review Rating—and additional behavioural variables could potentially provide further segmentation insights. Furthermore, the dataset represents a fixed collection of customer purchase records and does not provide real-time transaction data.

The current project can be further extended by incorporating additional analytical and machine learning techniques. Future work could include developing predictive models to forecast customer purchase behaviour, analysing customer lifetime value, and applying recommendation techniques to provide personalized product suggestions. The segmentation analysis could also be enhanced by incorporating additional customer behavioural features and comparing different clustering algorithms. Furthermore, the Streamlit dashboard could be extended with real-time or regularly updated datasets to support continuous monitoring of customer trends.

14. REFERENCES

- **Kaggle** – Customer Shopping Dataset
- Python Documentation
- Pandas Documentation
- Scikit-learn Documentation
- Streamlit Documentation
- Matplotlib Documentation
- Seaborn Documentation